Analog Pad Buffers for IHP SG13CMOS5L

Design notes: figures of merit, reference topologies, and a bandwidth budget

Christoph Maier

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1. Scope

Working notes toward a differential-input, analog-output pad driver (and its dual, a single-ended-pin to differential on-chip input buffer) for the IHP SG13CMOS5L open-source variant, in the context of a Chipalooza submission. These notes cover:

- which figures of merit are invariant under bias-current scaling, and are therefore the ones worth arguing about;
- what SG13CMOS5L removes relative to SG13G2, and how each omission constrains the topology choice;
- a reference bibliography of class-AB output stages;
- the relationship between the Monticelli and Hogervorst/Huijsing lineages;
- a bandwidth budget with conservative / reasonable / aggressive tiers.

Nothing here is silicon-verified. Numbers marked *estimate* are order-of-magnitude reasoning, not simulation results.

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2. Figures of merit, irrespective of bias current

If W and I are scaled together at fixed inversion level, then g_m, GBW and SR all scale $\propto I$ while noise power scales $\propto 1/I$. The quantities that survive that scaling are exactly the dimensionless ratios and the per-current groups.

2.1 Genuinely intensive (no current anywhere)

Quantity	Units	Comment
g_m/I_D	V^{-1}	Efficiency coordinate; everything else is downstream of where you sit on this curve
$(g_m/I_D)\cdot f_T$	Hz/V	The honest speed–efficiency invariant; peaks in moderate inversion, $IC \approx 0.1\ldots1$
$g_m r_o$	—	Intrinsic gain; sets static gain error $1/(\beta A_0)$. In 130 nm: $\approx 25\ldots30$ at $L_{\min}$, $100\ldots300$ at $L \approx 1$ µm
$I_{\text{out,peak}}/I_Q$	—	Class-AB current gain. Class A pins this at 1. Decent AB stages: 10–100
$V_{\text{out,pp}}/V_{\text{DD}}$	—	Swing efficiency, i.e. $\sum V_{\text{DSsat}}$ lost

Quantity	Units	Comment
$A_{VT} = V_{os}\sqrt{WL}$	mV·μm	Matching constant, $\approx 3 \ldots 5$ mV·μm thin-oxide; useful normalized form is $V_{os}/V_{out,pp}$
CMRR, PSRR	dB	At the frequency of interest
GBW/f_T	—	Technology utilization; usually humiliating, and usefully so
$\mathrm{SR}/(\mathrm{GBW} \cdot V_{step})$	—	$> 2\pi$ means settling is linear rather than slew-limited
$\eta = P_{load}/P_{supply}$	—	25 % ceiling for class A into a resistive load

2.2 Per-current groups

Invariant, but only comparable at equal C_L and equal swing:

- $\mathrm{FoM_{SS}} = \mathrm{GBW} \cdot C_L/I_{DD}$ [MHz·pF/mA]
- $\mathrm{FoM_{LS}} = \mathrm{SR} \cdot C_L/I_{DD}$ [V·pF/(μs·mA)]
- NEF / PEF, or the raw invariant $v^2_{n,\mathrm{in}} \cdot I_{DD}$
- HD3 or SFDR, quoted at fixed $V_{out,pp}/V_{DD}$ **and** fixed g_m/I_D — otherwise it measures the overdrive choice, not the topology

2.3 Pad-specific, and where such buffers actually fail

- **Transparency**: $C_{in}/C_{\text{tapped node}}$, plus kickback charge per unit input swing. For a monitor buffer on a low-current node this dominates everything else — the buffer that measures the DAC must not be the DAC's dominant load.
- **Stability across the load corner**: phase margin over C_L from ~5 pF (bare pad + ESD) to ~50 pF + coax. A single-number GBW is meaningless without the load-range envelope.
- $|Z_{out}(f)|$ and back-drive tolerance.
- Oxide field at the pad under ESD clamp turn-on.

2.4 The one that is not a ratio

Area. Offset and matching drag it in through $A_{VT}/\sqrt{WL}$, so any FoM table that omits area is quietly comparing a 40 μm^2 driver to a 4000 μm^2 one.

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3. What SG13CMOS5L takes away

SG13CMOS5L is not SG13G2 minus the HBTs. Per IHP's open-source page, it carries the SG13CMOS front end but has **no isolated NMOS, no MIM capacitor, four thin metals and a single 2 μm TopMetal**. Three of those hit an output driver directly.

3.1 No MIM capacitor

The compensation capacitor becomes a thick-oxide MOScap in accumulation or a MOM comb on M1–M4. With only four thin metals, MOM density is poor and the capacitor sits low in the stack where it couples to substrate. A MOScap is voltage-dependent even in accumulation.

Consequence: any topology whose stability depends on precise pole–zero cancellation — Miller with nulling resistor, nested Miller, active-feedback compensation — becomes a corner-analysis problem. This is the strongest single steer in the whole exercise:

> *Prefer **load-compensated single-stage** architectures, where the dominant pole is C_L at the pad and C_c is either absent or non-critical. Folded cascode or current-mirror OTA with a class-AB output — not a two-stage Miller amplifier.*

It also devalues the multistage-compensation literature (Leung & Mok damping-factor control, Ho/Mok active feedback), which assumes a well-characterized linear C_c.

3.2 No deep n-well / isolated NMOS

Every NMOS bulk is the global p-substrate, shared with whatever digital neighbour the shuttle places next door.

- NMOS source followers take full body effect with no escape ⇒ PMOS devices in their own n-well become the preferred followers and level shifters.
- Substrate-coupled noise enters the output stage's NMOS side unfiltered ⇒ keep the signal path PMOS-dominant where possible, and **carry the signal off-chip differentially rather than single-ended** if the measurement setup permits. If it must be single-ended at the pad, route a reference pad alongside as a pseudo-differential return and budget for the substrate noise that cannot be filtered.

3.3 One TopMetal, 2 µm, no TM2

Supply distribution to the output stage and the ESD return path are both thinner than in SG13G2. This caps peak drive current and makes IR drop across the pad ring a real term in the swing budget. It also removes a shielding layer: with four thin metals, one must be spent on a shield plane over the input pair, which is expensive when M4 is also the best MOM plate and the best local supply mesh.

3.4 Revised FoM weighting

Given the above, the discriminating figures for this process are:

1. swing efficiency $V_{\text{out,pp}}/\sum V_{\text{DSsat}}$ on the 3.3 V thick-oxide rail;
2. phase margin across the full C_L envelope **with C_c at its ±30 % MOScap corner**;
3. PSRR and substrate-coupling rejection — a first-class spec here, not an afterthought;
4. area, since MOM/MOScap compensation is area-hungry and shuttle slots are small.

g_m/I_D and intrinsic gain still set the design point but no longer differentiate between candidates; the compensation and isolation constraints do.

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4. Reference designs — class-AB output stages

#	Design / paper	Venue	Link
1	Monticelli, *A quad CMOS single-supply op amp with rail-to-rail output swing* — the floating-battery AB mesh	JSSC SC-21(6), 1986, 1026–1034	PDF (Iowa State) · doi:10.1109/JSSC.1986.1052645
2	Carvajal, Ramírez-Angulo et al., *The flipped voltage follower* — FVF/DFVF taxonomy	TCAS-I 52, 2005, 1276–1291	doi:10.1109/TCSI.2005.851387
3	Sawigun & Demosthenous, *A compact rail-to-rail class-AB CMOS buffer with slew-rate enhancement*	TCAS-II, 2012	IEEE Xplore
4	*High-speed rail-to-rail class-AB buffer with compact adaptive biasing for FPD*	MDPI Electronics 9(12), 2020	open access
5	Ramírez-Angulo / Carvajal et al., *Class-AB output stages with accurate quiescent current control by dynamic biasing*	Analog Integr. Circ. Sig. Process.	Springer
6	*Low-voltage class-AB output stages using floating capacitors* — the sub-2V_{GS} alternative to Monticelli	Electron. Lett. lineage	PDF
7	Hogervorst et al., *Rail-to-rail constant-g_m input stage and class-AB output stage*	Analog Integr. Circ. Sig. Process.	Springer
8	*Rail-to-rail low-power high-slew-rate CMOS analogue buffer* — <32 fF input capacitance	—	Academia
9	*Rail-to-rail high-speed class-AB CMOS buffer with enhanced slew rate* — 1 nF at 3.5 µA I_Q	—	ResearchGate

#	Design / paper	Venue	Link
10	López-Martín et al., *The Flipped Voltage Follower: Theory and Applications*	Springer LNEE	chapter

Bibliography only (link not verified; all appear in the reference lists of the above):

- F. You, S. H. K. Embabi, E. Sánchez-Sinencio, "Low-voltage class-AB buffers with quiescent current control," *JSSC* 33(6), 1998, 915–920.
- M. de Langen, J. H. Huijsing, *JSSC* 33(10), 1998, 1482–1496.
- G. Giustolisi, G. Palmisano, "1.2-V op-amp with a dynamically biased output stage," *JSSC* 35(4), 2000, 632–636.
- M. D. Pardoen, M. G. Degrauwe, "A rail-to-rail CMOS input/output power amplifier," *JSSC* 25(4), 1990, 501–504.
- J. A. Fisher, R. Koch, "A highly linear CMOS buffer amplifier," *JSSC* 22, 1987, 330–334.
- R. Hogervorst, J. P. Tero, R. G. H. Eschauzier, J. H. Huijsing, "A compact power-efficient 3 V CMOS rail-to-rail input/output operational amplifier for VLSI cell libraries," *JSSC* 29(12), Dec. 1994, 1505–1513.

4.1 Reading order for this process

Items 1 and 6 should be read against each other: Monticelli's mesh costs more than $2V_{GS}$ of headroom but needs no capacitor, whereas the floating-capacitor variants buy headroom back by spending exactly the component SG13CMOS5L does not provide. Items 3 and 4 are the useful pair on slew enhancement. Item 2 is the compactness reference whose offset and PSRR must be characterized independently, since the papers largely do not.

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5. Monticelli vs. Hogervorst & Huijsing — different axes

They are not the same topology, and the confusion is worth dissolving because the two names sit on **different axes**.

Monticelli (1986) is about a class-AB *output* stage plus quiescent-current control. Its input stage is a conventional single pair: the abstract claims an output swing to either rail together with an input common-mode range that includes ground — ground-sensing, *not* rail-to-rail input.

Hogervorst & Huijsing's headline contribution is precisely what Monticelli lacks: the **constant-g_m rail-to-rail input stage**, two complementary pairs with tail-current steering so $g_{m,\text{tot}}$ stays flat as the common mode sweeps. They publish output stages too, but the usual pairing in the literature is *Hogervorst input + some floating class-AB output*, and that output can perfectly well be Monticelli's mesh. **They compose rather than compete.**

5.1 Where they do overlap

Both output stages belong to Huijsing's **feedforward** class-AB control family: a floating voltage source (translinear loop) between the two output gates. This is distinct from **feedback** control with an explicit minimum-current selector — the de Langen & Huijsing branch (*JSSC* 1998), which senses the smaller of the two output currents and regulates it, giving tighter I_Q over corners at the cost of a loop that can ring.

So Monticelli and the Hogervorst compact output stage are cousins; the Delft feedback-controlled stages are a different animal.

5.2 Two practical differences

Headroom. Monticelli's mesh needs roughly $2V_{GS} + 2V_{\text{DSsat}}$ of stack. That is exactly the criticism levelled by the floating-capacitor school, whose stated advantage is accurate quiescent control at supplies near one threshold, where Monticelli's requires more than $2V_{GS}$. On a 3.3 V thick-oxide rail with $V_T \approx 0.5\ldots0.7$ V this is affordable; on the 1.2 V core it is not.

Drive asymmetry. The mesh is not symmetric in the small-signal path. One output device's gate is driven directly by the cascode current; the complementary device is driven *through* the mesh, which behaves as an additional cascode stage with a pole set by $1/g_m$ of the mesh device and the capacitance at that node — contributing a non-dominant pole and extra delay on that path only. This shows up as unequal rising/falling settling, and it interacts badly with a compensation capacitor of uncertain value. Treat that pole's location as a corner variable and simulate it explicitly rather than trusting a nominal phase margin.

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6. Hogervorst et al. 1994 as a donor design

R. Hogervorst, J. P. Tero, R. G. H. Eschauzier, J. H. Huijsing, "A Compact Power-Efficient 3 V CMOS Rail-to-Rail Input/Output Operational Amplifier for VLSI Cell Libraries," JSSC 29(12), Dec. 1994, 1505–1513.

Verdict: usable as an **output-stage donor, not as a cell to copy**.

6.1 Keep — the output stage and its biasing (Figs. 5, 8, 11)

The floating class-AB control M_{19}/M_{20} shifted *into* the folded-cascode summing circuit means the AB control contributes neither noise nor offset: the two bias sources I_{b6}/I_{b7} that would otherwise sit in parallel with the cascodes M_{14}/M_{16} are eliminated outright. The floating current source M_{27}–M_{28}, built with the *same* translinear structure as the AB control so that its supply dependence cancels the control's, yields an I_Q that does not walk with V_{DD}. A real solution to a real problem, at a cost of two transistors.

Table I of that paper was measured at exactly 3.3 V — the same rail as IOVDD on the 5L shuttle — with two stacked V_{GS} in the output stage, which on IHP thick-oxide devices ($V_T \approx 0.5\ldots0.7$ V) leaves room.

6.2 Discard — the constant-g_m rail-to-rail input apparatus (Section II)

For a differential *on-chip* input at fixed, known common mode, everything in their Figs. 1–4 is pure cost: the complementary pairs, the two current switches, the $3\times$ mirrors M_6–M_7 and M_9–M_{10}, plus M_{29}–M_{31} whose only job is to keep the positive-feedback loop from arming below 2.9 V.

What it buys in defects, from their own Table I and Section VI:

- CMRR collapses to **43 dB** in the take-over ranges, against 70 dB elsewhere;
- offset shifts about **2 mV per take-over range**;
- slew rate changes by a **factor of two** depending where the common mode sits ($2 \to 4$ V/µs Miller; $4 \to 8$ V/µs cascoded-Miller).

A single PMOS pair at fixed common mode gives constant g_m, constant SR, no take-over artifacts, better CMRR, a bulk-tied-to-source device in its own n-well (which matters with no deep n-well available), and roughly 40 % of the area back.

6.3 The porting obstacle — C_{M1}/C_{M2}

They are Miller capacitors tied from V_o to the output-device gates, so the **full output swing stands across them**. With MIM that is free; in 5L it means MOM fingers on four thin metals, or a thick-oxide MOScap that will traverse accumulation-to-depletion as the output swings — a compensation capacitor that changes value with the signal it is compensating.

Two mitigations available from the paper itself:

1. The plain-Miller version has **66° phase margin** versus **53°** for cascoded-Miller. Take the Fig. 13 variant and spend the margin on capacitor uncertainty rather than on bandwidth.
2. They note cascoded-Miller can peak at high output current and recommend plain Miller above ~3 mA — which points the same way for a driver.

6.4 Two numbers that will not survive the port

- **Gain.** Their 85 dB came from 1 µm devices. On 130 nm thick-oxide with the same cascoding, budget 60–70 dB (*estimate*); their own pointer to Bult & Geelen gain boosting on M_{14}/M_{16} becomes less optional.
- **Load.** 10 kΩ ∥ 10 pF at 3 mA peak is a *chip-level* buffer. A metre of coax plus a scope front end is 100 pF and change. Scaling the output devices for that raises $C_{GS,\mathrm{out}}$, drags the output pole down, and forces C_M up — straight back into the capacitor problem. Size that loop before committing.

Their figure of merit, $F = B/P_{\mathrm{sup}}$ at $C_L = 10$ pF, is supply-referred rather than current-referred; convert to $\mathrm{GBW} \cdot C_L/I_{\mathrm{DD}}$ to compare against anything modern.

6.5 Incidental

The Fig. 6 → 8 → 9 → 10 → 11 progression, in which each figure repairs one named defect of its predecessor, is unusually clean design-rationale writing. If the shuttle wants documentation rather than only a GDS, that structure is worth reusing as a template.

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7. Bandwidth budget

7.1 The load sets everything on the output side

For a load-compensated single stage,

$$\mathrm{GBW} = \frac{g_m}{2\pi C_L}$$

C_L is not the pad. It is pad (a few hundred fF) + ESD (~1 pF) + bond wire + PCB trace + whatever instrument is attached. A 10× scope probe alone is 10–15 pF. **Budget $C_L \approx 15$ pF** as the honest bench number, and 50–100 pF if anyone ever hangs un-terminated coax on it.

At $C_L = 15$ pF:

Target GBW	Required g_m	I_D at $g_m/I_D = 10\ \mathrm{V}^{-1}$
10 MHz	0.94 mS	≈ 95 µA
100 MHz	9.4 mS	≈ 1 mA

Current is not the wall. **The non-dominant poles are.** A thick-oxide 3.3 V device at $L \approx 1$ µm in moderate inversion has f_T of perhaps 1–3 GHz (*estimate*), and the mirror and cascode poles will land in the low hundreds of MHz. Pushing GBW to 100 MHz puts the unity-gain crossing uncomfortably close to them — while the compensation capacitor's value is uncertain because there is no MIM.

7.2 Slew usually binds first

Full-power bandwidth requires $\mathrm{SR} = 2\pi f V_{\mathrm{pk}}$, i.e. peak current $I = C_L \cdot \mathrm{SR}$. For 1 V peak at 10 MHz into 15 pF that is ≈ 0.94 mA of peak drive — comfortable for a class-AB stage running 50 µA quiescent. At 100 MHz it is 9.4 mA, which on a single 2 µm TopMetal supply route starts consuming the swing budget in IR drop.

Quote small-signal GBW and full-power bandwidth separately. They are not the same spec, and the gap between them is the entire point of class AB.

7.3 Calibration tiers — output buffer (differential in → single-ended pin)

Tier	GBW	FPBW at 1 $\mathrm{V_{pk}}$	Risk
Conservative	1–5 MHz	~1–2 MHz	first silicon works, no drama
Reasonable	**10–30 MHz**	**5–10 MHz**	sweet spot; poles comfortably clear
Aggressive	50–100 MHz	20 MHz	pole management becomes the whole design; MOScap C_c uncertainty acute
Don't	> 200 MHz	—	stop calling it a buffer

Above roughly 100 MHz through a wire-bonded pad into an unknown board, the problem stops being "amplifier" and becomes "transmission line": an impedance-matched 50 Ω driver is wanted, the load turns resistive, and power rises by an order of magnitude. Different project.

7.4 The input buffer is an easier problem

Single-ended pin $\rightarrow$ differential on-chip drives only internal capacitance — call it 200–500 fF. 50 MHz needs under 100 µS of g_m. The limits here are the ESD structure's series RC, the input pair's own matching and noise, and the fact that deliberate band-limiting is usually *wanted* for anti-aliasing and noise.

50–100 MHz is comfortable; 20 MHz is conservative. The asymmetry between the two directions is real — roughly a decade of headroom.

7.5 Two arguments for staying low

- Noise scales as $\sqrt{\mathrm{BW}}$. If the path characterizes anything quiet, excess bandwidth is a direct loss of resolution.
- Keeping GBW modest lets C_L be the dominant pole and lets the Miller capacitor be dropped entirely — the cleanest escape from the no-MIM problem of §3.1.

7.6 Working stake

20 MHz out, 50 MHz in. With the caveat that the right answer depends on what is being observed: DC bias characterization needs 1 MHz; observing a settling transient needs whatever bandwidth resolves the settling constant of interest.

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8. FPBW — definition

Full-Power Bandwidth: the highest frequency at which the amplifier still delivers its *full rated output swing* without the waveform being distorted by slew-rate limiting.

The derivation is one line. For $V_{\text{out}} = V_{\text{pk}} \sin(2\pi f t)$, the maximum rate of change occurs at the zero crossing:

$$\left.\frac{dV}{dt}\right|_{\max} = 2\pi f V_{\text{pk}}$$

The amplifier can only supply dV/dt up to its slew rate. Setting them equal:

$$\boxed{\mathrm{FPBW} = \frac{\mathrm{SR}}{2\pi V_{\text{pk}}}}$$

Above that frequency the output stops being a sine and becomes a triangle wave: the amplifier is charging C_L with all the current it has and cannot turn around fast enough. This is a hard, ugly, large-signal nonlinearity, not a gentle roll-off.

8.1 Why it is separate from GBW

GBW is a **small-signal** spec — how fast the amplifier responds to a wiggle small enough that everything stays linear. FPBW is **large-signal** — how fast it responds when asked for the whole swing.

A class-A stage with fixed tail current I has $\mathrm{SR} = I/C_L$, and its FPBW can be a factor of ten or more below its GBW. It then looks fast on a Bode plot and turns a 1 V sine into a triangle. Closing that gap is exactly what class AB is for: on a large step the output devices deliver many times the quiescent current, so SR is no longer pinned to I_Q.

The ratio $\mathrm{SR}/(\mathrm{GBW} \cdot V_{\text{step}})$ from §2.1 is the same idea in dimensionless form. Above about 2π, the amplifier settles by linear exponential decay and the FPBW spec does not constrain; below it, slewing dominates the settling time and GBW is a fiction.

8.2 Consistency check

1 V peak at 10 MHz needs $\mathrm{SR} = 2\pi \times 10^7 \times 1 \approx 63$ V/µs; into 15 pF, $I = C \cdot \mathrm{SR} \approx 0.94$ mA — the same figure as §7.1. The peak-current and FPBW framings are one calculation wearing two hats.

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9. Open questions before committing

1. Does `ihp-sg13cmos5l` ship an **analog pad cell** — an unbuffered pad with a characterized ESD network and known parasitic capacitance — or only digital I/O cells? If digital-only, the choice is between designing an ESD structure (probably not sign-off-able on a shared shuttle) and living with a digital pad's diode clamps in the analog path.
2. What does the Chipalooza floorplan allow for **pad allocation**, and is an analog supply domain separate from IOVDD available at all?
3. Is the far end of the link a **differential receiver**? If so, do not collapse to single-ended on chip (§3.2).
4. What is the actual C_L envelope of the intended bench setup? Everything in §7 hangs on it.

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10. Provenance and caveats

- The SG13CMOS5L device/BEOL inventory is taken from IHP's open-source request page, which described the open PDK as under development for early 2026. Verify against the `IHP-GmbH/ihp-sg13cmos5l` repository rather than the marketing page.
- Items 1–10 in §4 have verified links. The bibliography-only list does not; those citations come from the reference lists of the linked papers.
- The Monticelli-vs-Huijsing feedforward/feedback taxonomy in §5.1 is asserted from secondary sources. Huijsing's *Operational Amplifier Theory and Design* carries the canonical version and should be checked before the distinction carries weight in a design justification.
- Every number marked *estimate* is order-of-magnitude reasoning. Nothing here has been simulated in SG13CMOS5L.